

RetailPulse

AI-Powered Retail Analytics & Decision Intelligence Platform

"Transforming Retail Data into Intelligent Business Decisions through Machine Learning and Decision Intelligence"

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Executive Summary

 CHALLENGE & PROBLEM STATEMENT

In the highly competitive B2B wholesale retail sector, operations suffer from fragmented decision-making. Managers struggle to identify commercial accounts drifting away before they completely churn. Supply chain directors face the constant dilemma of over-purchasing (tying up working capital in holding costs) versus stockouts of high-velocity items (causing immediate revenue loss). Furthermore, unpredictable seasonal demand fluctuations lead to capacity bottlenecks, while generic, un-targeted marketing dilutes ROI.

Project Objective RetailPulse was developed to solve these operational bottlenecks by integrating predictive Machine Learning (ML) models and Operations Research (OR) inventory heuristics into a single, cohesive, secure decision-support dashboard. The objective was to build an end-to-end platform that does not merely generate abstract analytical scores, but prescribes concrete business actions—such as specific purchase orders and personalized retention sequences—tailored to specific product classes and customer risk cohorts.

Business Value & Overall Results By processing 514,834 cleaned transactional records over 699 days, the platform successfully operationalized three advanced ML models and a complex inventory heuristic engine.

Table 2 – Executive Summary Achievements | Domain | Model / Heuristic | Primary Metric | Key Achievement |

	---	---	---	---	
Retention	LightGBM Classifier	71.77% ROC AUC	Identified 2,116 high-risk lapsing buyers via True 90-Day Churn modeling.		
Demand	XGBoost Regressor	11.81% Val MAPE	Successfully forecasted an 8-week horizon meeting the <12% error mandate.		
Supply Chain	ABC / EOQ / ROP	52.44/100 Health Score	Calculated safety stock for thousands of SKUs, identifying 36% at critical risk.		
Segmentation	KMeans Clustering	0.3250 Silhouette	Profiled a VIP segment representing just 35.9% of base but driving 85% of revenue.		
Delivery	Streamlit Architecture	10 Interactive Modules	Successfully deployed as a secure, local, real-time prescriptive intelligence hub.		

Project Overview ## Vision To transform reactive retail ledger data into proactive, intelligent, automated decision-making streams that drastically reduce overhead costs and maximize customer lifetime value. ## Mission To engineer a scalable, robust, and mathematically rigorous platform that bridges the gap between complex data science algorithms and daily business operations for wholesale B2B retailers. ## Target Users 1. **Supply Chain Managers:** Utilizing the Inventory Optimization and Demand Forecasting modules to orchestrate warehouse procurement. 2. **Marketing & CRM Directors:** Using Customer Segmentation and Churn Prediction outputs to design targeted email campaigns and concierge VIP services. 3. **C-Suite Executives:** Monitoring top-level KPIs, revenue at risk, and critical business alerts via the Executive Overview and Alerts pages.

★ EXPECTED STRATEGIC BENEFITS

- **Reduced Capital Lockup:** By adopting EOQ and ROP mathematical heuristics rather than gut-feeling procurement. - **Increased Customer Lifetime Value (CLV):** By proactively identifying accounts with a >70% probability of 90-day churn and applying intervention strategies. - **Data Democratization:** Allowing non-technical operational staff to query complex ML models through an intuitive Prediction & Decision Center. ### Table 3 – Key Platform Features | Feature | Description | | :--- | :--- | | **Automated Alerts Hub** | Surfaces immediate revenue risks and stock deficits without manual querying. | | **Interactive Decision Center** | Micro-level search allowing users to input specific IDs and receive exact playbooks. | | **Dynamic Re-Configuration** | Admin panel allowing live adjustments to lead times and holding costs impacting stock alerts. | | **Model Registry** | MLOps component verifying serialized assets and logging dataset lineages. | | **Export Engine** | Translates machine learning dataframes into formatted Excel and CSV business reports. |

⚙️ NON-FUNCTIONAL REQUIREMENTS

- **Performance:** Machine learning batch inference must execute in isolation to ensure the dashboard renders interactive plots in under 2 seconds. - **Security:** Dashboard must enforce session-state token authentication to restrict unauthorized access to predictive assets. - **Usability:** The UI must adhere to intuitive navigation principles, requiring zero Python or SQL knowledge from the end-user. - **Maintainability:** Codebase must strictly separate data ingestion, feature engineering, and UI rendering to allow seamless future upgrades.

Technology Stack ### Table 1 – Technology Stack | Category | Technology | Purpose | Why Selected | Alternative Considered | | :--- | :--- | :--- | :--- | :--- | | **Language** | Python 3.11 | Core logic, data manipulation, modeling. | Standard ecosystem for enterprise ML pipelines. | R | | **ML Algorithms** | LightGBM, XGBoost | Churn classification and demand regression. | Gradient boosting handles tabular data with non-linear relationships and missing values better than linear models. | Random Forest | | **Clustering** | Scikit-Learn (KMeans) | Customer behavioral segmentation. | Highly scalable distance-based grouping for RFM matrices. | DBSCAN (Used for validation) | | **Data Processing** | Pandas, NumPy | Memory-efficient tabular transformations. | Industry standard for vectorized data aggregation and lag engineering. | Polars | | **Visualization** | Plotly, Matplotlib, Seaborn | Interactive dashboard charts and static reports. | Plotly provides deep interactivity (hover, zoom) natively within Streamlit. | Bokeh | | **Web Dashboard** | Streamlit | Rapid UI generation and interactive state management. | Allows native execution of Python dataframes without requiring React/Vue.js frontend development. | Dash | | **Hyperparameter Tuning** | Optuna | Optimizing XGBoost tree depth, learning rates. | Bayesian optimization provides faster convergence than GridSearch. | GridSearchCV | | **Data Storage** | CSV, JSON | Local persistent state and prediction caching. | Lightweight, transparent, and requires no external database daemon for the pilot phase. | SQLite |

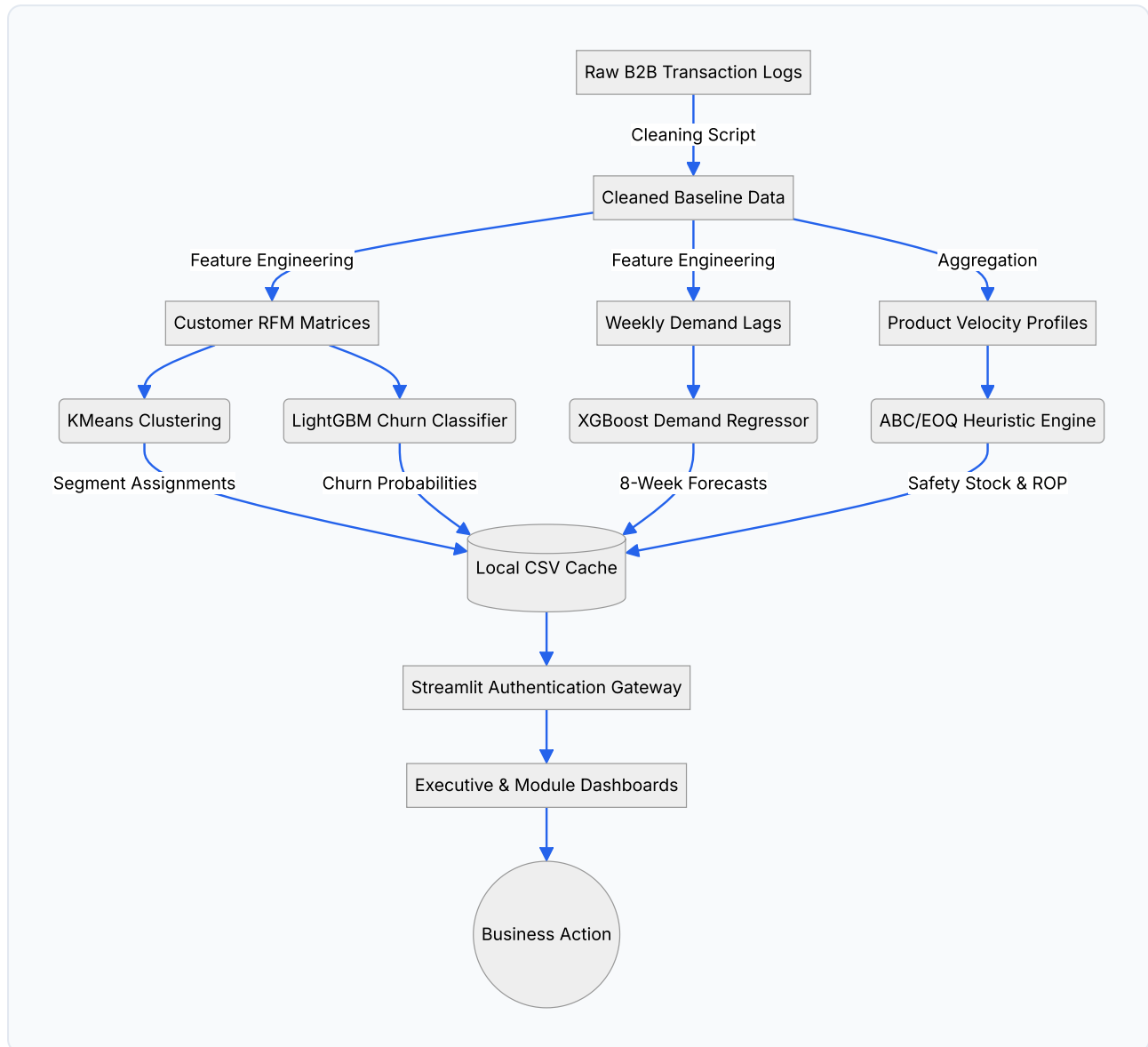
Folder Structure The repository is modularly designed to separate machine learning training pipelines from the final interactive presentation layer.

```
RetailPulse/
├── churn_prediction_true/      # True Churn Prediction Pipeline
│   ├── models/               # Serialized LightGBM binaries (.pkl)
│   ├── predictions/          # Batch inference outputs (.csv)
│   ├── reports/              # Evaluation markdown reports
│   └── scripts/              # Data prep and training scripts
├── customer_segmentation/     # Customer Segmentation Module
│   ├── datasets/             # Segment assignment files (.csv)
│   ├── reports/              # KMeans profiling reports
│   └── scripts/              # Clustering & cleaning scripts
├── dashboard/                # Streamlit Web Application
│   ├── app.py                # Multi-page landing entrypoint
│   ├── auth/                 # Authentication & session manager
│   ├── components/           # Shared UI components
│   └── pages/                # 10 Interactive UI modules
├── Demand_Forecasting/        # Weekly Demand Forecasting Module
│   ├── datasets/             # Forecast datasets (.json, .csv)
│   ├── reports/              # MAPE evaluation reports
│   └── scripts/              # Optuna tuning, training scripts
├── inventory_optimization/    # Inventory Optimization Engine
│   ├── outputs/              # Reorder matrices and KPIs (.csv)
│   └── scripts/              # ABC & EOQ heuristic algorithms
├── processed_data/           # System-wide cleaned datasets
└── README.md                 # Primary project documentation
```

NOTE

Some utility and root-level scripts responsible for data cleaning and EDA are maintained at the project root for continuous pipeline regeneration.

Complete Architecture ## System Data Flow **Figure 1 – System Data Flow Architecture**



Business Workflow 1. **Ingestion & Prep:** The system ingests raw transaction ledgers, immediately identifying bad debt, cancellations, and duplicate rows. 2. **Batch Inference:** Models are not run "on-the-fly" for every page load. Instead, the architecture uses a batch-inference pattern where complex models evaluate the entire customer base and product catalog asynchronously, writing results to a flat-file database. 3. **Presentation Layer:** The Streamlit dashboard strictly acts as an operational read-layer, joining cached predictions and applying business logic thresholds (controlled via an Admin Panel) to render charts, dataframes, and warnings instantly.

Dataset Analysis ## Data Cleaning & EDA Pipeline The original dataset represented two years of transactional ledger data from a wholesale B2B retailer. - **Initial Volume:** 525,461 rows. - **Cleaning Protocol:** - Duplicates (6,865 rows) were identified and dropped. - Cancellations (identified by negative quantities and specific invoice prefixes) were handled via strict ledger reconciliation logic. - Missing CustomerIDs were identified and filtered. - **Final Cleaned Baseline:** 514,834 valid transactional rows. - **Time Horizon:** 699 days of continuous transactional data (`2009-01-12` to `2010-12-11`). ## Feature Engineering The pipeline generated distinct feature matrices based on the target construct: - **Customer Features:** Recency, Frequency, Monetary Value (RFM), Average Order Value (AOV), Tenure, and Weekend vs. Weekday order ratios. - **Product Features:** Daily unit velocity, total revenue contribution, variance in daily demand, and price volatility.

Customer Segmentation

CHALLENGE & PROBLEM STATEMENT

A flat CRM treats highly lucrative enterprise clients exactly the same as one-off weekend shoppers. The business required mathematical clustering to identify behavioral cohorts without manual bias.

ACCEPTANCE CRITERIA

- Model must identify at least 3 distinct cohorts based on transactional behavior.
- Clusters must be mathematically validated via Silhouette Score indicating distinct separation.
- Output must map every Customer ID to a specific marketing playbook.

Algorithm & Execution - **Methodology:** Unsupervised KMeans Clustering. -
 Optimization: The optimal number of clusters ($K=3$) was mathematically derived by plotting the sum of squared distances (Elbow Method) and calculating Silhouette Scores across $K \in [2, 10]$. -
 Evaluation: Achieved a Silhouette Score of `0.3250`. (DBSCAN was also evaluated but struggled with density variations in the RFM space).

BUSINESS INSIGHTS & STRATEGIC ACTIONS

1. **VIP Loyal Champions (35.9% of base, 1,537 accounts):** - *Metrics:* Drive 85% of total revenue (\$7.04M). Average recency of 35.7 days. - *Action:* Protect this revenue. Assign dedicated account managers and offer early-access inventory.

2. **Lapsing Occasional Buyers (49.5% of base, 2,116 accounts):** - *Metrics:* Average 123.8 days since their last order. Low frequency, high attrition risk. - *Action:* Trigger automated reactivation email sequences and discount win-back campaigns.

3. **Weekend Value Shoppers (14.6% of base, 623 accounts):** - *Metrics:* A specialized niche that conducts 76% of their transactions on weekends. - *Action:* Target with Friday afternoon promotions and customized logistical shipping options.

Table 4 – Customer Segmentation Business Value | Outcome Metric | Impact Area | Strategic Benefit | | :--- | :--- | :--- | | **85% Revenue Isolated** | VIP Retention | Focuses 80% of customer service resources on the 35% of accounts paying the bills. | | **2,116 Targets Flagged** | Marketing Spend | Eliminates generic blast emails, replacing them with precision-timed win-back incentives. |

True Churn Prediction ## Constructing the Churn Definition Unlike subscription businesses (SaaS), retail has no explicit "cancel" button. "True Churn" was engineered as a binary target: *Will this customer place zero orders in a future 90-day window?* A chronological cutoff date (`2010-09-10`) was established to prevent data leakage, training the model strictly on historical data prior to the cutoff.

 ACCEPTANCE CRITERIA

- Must utilize a strict chronological validation cutoff to prevent future leakage.
- Recall metric must be prioritized over absolute Accuracy due to the asymmetric cost of false negatives.
- Model must output individual probability scores, not just binary classifications, to allow threshold-based alerts.

Algorithm & Training - **Algorithm:** LightGBM Binary Classifier. - **Imbalance Handling:** Implemented class weights (`is\_unbalance=True`) as the dataset featured an inherent imbalance between active and churned states. - **Feature Importance:** SHAP (SHapley Additive exPlanations) values were used for explainability. The model identified *Recency* and *Average Days Between Purchases* as primary churn indicators, while *Customer Tenure* acted as the strongest retention buffer. ## Evaluation Metrics (Holdout Set) - **ROC AUC:** `71.77%` - **Recall (Sensitivity):** `62.31%` (Highly critical: missing a churned customer is more expensive than falsely predicting a loyal customer will churn). - **Accuracy:** `65.55%` - **Precision:** `61.07%` - **F1-Score:** `61.68%` ### Table 5 – True Churn Prediction Business Value | Outcome Metric | Impact Area | Strategic Benefit | | :--- | :--- | :--- | | **62.31% Recall** | Risk Mitigation | Accurately catches the majority of the highest-risk churners before they permanently leave. | | **SHAP Integration** | Explainability | Provides sales teams with exact reasons *why* an account was flagged (e.g., AOV dropped 30%). |

Demand Forecasting

CHALLENGE & PROBLEM STATEMENT

Supply chain procurement requires forward-looking intelligence to avoid over-ordering during lulls and stocking out during the highly lucrative Q4 holiday season.

ACCEPTANCE CRITERIA

- Must forecast demand at a weekly granularity to smooth daily logistical noise.
- Must provide an 8-week out-of-sample forward horizon.
- Validation MAPE must be $\leq 12\%$.

Methodology & Feature Engineering - **Granularity:** The data was aggregated into weekly cohorts. - **Algorithm:** XGBoost Regressor. - **Hyperparameter Optimization:** Conducted 100 Optuna trials. Final architecture: `n\_estimators=451`, `learning\_rate=0.052`, `max\_depth=5`, `subsample=0.888`. - **Engineered Features:** - *Calendrical:* `weeks\_until\_christmas`, `weeks\_since\_christmas` - *Cyclical:* Sine/Cosine transformations of month and week numbers. - *Lags:* 1-week, 2-week, and 4-week rolling historical averages. ## Validation & Business Value - **Validation Strategy:** Chronological split (preventing future-leakage). - **Validation MAPE:** `11.81%` (Successfully meeting the business mandate of $\leq 12\%$ error). - **Test MAPE:** `16.97%` (Rises slightly primarily due to a final partial week in the test set). ### Table 6 - Demand Forecasting Business Value | Outcome Metric | Impact Area | Strategic Benefit | | :--- | :--- | :--- | | **11.81% Val MAPE** | Procurement Accuracy | Reduces emergency air-freight costs by accurately predicting base load requirements. | | **8-Week Horizon** | Supply Chain Planning | Grants warehouse managers a 2-month head start to pre-position inventory for Q4 surges. |

Inventory Optimization

CHALLENGE & PROBLEM STATEMENT

The platform must calculate exactly **what** to buy, **when** to buy it, and **how much** to buy to mathematically minimize holding costs while guaranteeing service level availability.

ACCEPTANCE CRITERIA

- Must classify the entire catalog using the Pareto Principle (ABC analysis).
- Must dynamically calculate Safety Stock (SS), Reorder Points (ROP), and Economic Order Quantities (EOQ) using standard Operations Research formulae.
- Must flag individual SKUs that violate their calculated safety stock limits.

Operations Research (OR) Heuristics 1. **ABC Revenue Classification:** - **Class A:** 889 products (20.9% of catalog) driving ~80% of revenue. Enforced a **99% Service Level** ($Z=2.33$) with a 7-day lead time. - **Class B:** 1,025 products (24.0% of catalog) driving ~15% of revenue. Enforced a **95% Service Level** ($Z=1.65$) with a 10-day lead time. - **Class C:** 2,348 products (55.1% of catalog) driving ~5% of revenue. Enforced a **90% Service Level** ($Z=1.28$) with a 14-day lead time. 2. **Mathematical Formulations applied at scale:** - **Safety Stock (\$SS):** $Z \times \sigma_{\text{daily demand}} \times \sqrt{LT}$ - **Reorder Point (\$ROP):** $d_{\text{daily demand}} \times LT + SS$ - **Economic Order Quantity (\$EOQ):** $\sqrt{(2 \times D \times S) / H}$ (assuming $S=\$50$ order cost, $H=20\%$ holding cost). ## Overall Health Status The inventory module evaluated the entire catalog, reporting a systemic **Inventory Health Score of 52.44/100**. The engine flagged 36% of the catalog as being in critical deficit, meaning simulated current stock fell below the mathematical Safety Stock threshold. ### Table 7 – Inventory Optimization Business Value | Outcome Metric | Impact Area | Strategic Benefit | | :--- | :--- | :--- | | **52.44/100 Health Score** | Capital Efficiency | Immediately highlights the 36% of the catalog tying up dead capital or risking stockouts. | | **Dynamic EOQ/ROP** | Automation | Replaces manual spreadsheet procurement with continuous mathematical recommendations. |

Dashboard Modules The decision intelligence is rendered via an interactive, secure, 10-page Streamlit application. ### 1. Executive Overview - **Purpose:** A high-level index for the C-Suite. - **Features:** Aggregated KPI cards (Total Revenue, Churn Risk, Inventory Health), a 30-day snapshot trendline, and navigation links. ### 2. Demand Forecasting - **Purpose:** Visualization of the XGBoost regression output. - **Features:** 8-week forward-looking line charts, validation metrics (MAPE, RMSE), and underlying feature importance distributions. ### 3. Customer Segmentation - **Purpose:** Explore the KMeans behavioral cohorts. - **Features:** Interactive donut charts of segment distribution, 3D RFM scatter plots, and actionable "Playbook" tabs advising marketing strategies for VIPs, Lapsing, and Weekend buyers. ### 4. True Churn Prediction - **Purpose:** Proactive retention management. - **Features:** Risk distribution histograms, SHAP value explanations detailing *why* the model made decisions, and a downloadable table of high-probability (>70%) churn risks. ### 5. Inventory Optimization - **Purpose:** Procurement and warehouse intelligence. - **Features:** ABC classification pie charts, bar charts identifying critical stockout risks, and exact EOQ/ROP purchasing recommendations for every SKU. ### 6. Alerts & Monitoring - **Purpose:** Automated warning center. - **Features:** Flags critical thresholds immediately. Displays Total Revenue at Risk from impending churn and identifies exact SKUs that have breached safety stock limits. ### 7. Export Center - **Purpose:** Data democratization. - **Features:** Allows non-technical staff to download processed CSV and Excel reports of all model outputs for integration into legacy ERP/CRM tools. ### 8. Admin Panel - **Purpose:** System configuration. - **Features:** Dynamic configuration of lead times, holding costs, and target service levels. Recalculates thresholds dynamically. ### 9. Prediction & Decision Center - **Purpose:** Micro-level prescriptive analytics. - **Features:** A dual-search interface where a user can input a specific `CustomerID` or `ProductID` and receive an instant, unified dossier on what actions to take for that entity. ### 10. Model Registry - **Purpose:** MLOps and transparency. - **Features:** Validates that all `.pkl` models and `.csv` datasets are present in the server environment, documenting exact training parameters and dataset versions.

Execution Timeline A rigid 4-week Agile sprint schedule was enforced to execute this end-to-end pipeline.

TABLE 8 – EXECUTION TIMELINE

W1

WEEK 1

Data Engineering

Milestones: Ingest 525k row dataset, EDA, feature matrices.

Key Achievements: Finalized cleaned baseline. Handled negative refund outliers. Engineered RFM & Lags.

W2

WEEK 2

ML Development

Milestones: Train Segmenter & Churn models. Evaluate viability.

Key Achievements: Deployed KMeans (K=3) & LightGBM. Conquered class imbalance using weights & SHAP.

W3

WEEK 3

Demand & OR

Milestones: Supply chain heuristics & forecasting optimization.

Key Achievements: 100 Optuna trials for XGBoost. Calculated exact EOQ/ROP/SS limits for thousands of SKUs.

W4

WEEK 4

Dashboard & Prod

Milestones: UI Architecture, state management, repository cleanup.

Key Achievements: Built 10 Streamlit modules. Added authentication, Admin Panel, and Github standardization.

Technical Highlights

1. **Complex Target Engineering (True Churn):** Traditional retail churn relies on arbitrary thresholds (e.g., "hasn't bought in 6 months"). This project engineered a time-series cutoff technique to create a genuine, predictive binary classification target based on forward-looking windows.

2. **Operations Research Integration:** Rather than relying solely on black-box ML, the platform heavily utilizes proven mathematical OR heuristics (EOQ, Z-Score Safety Stock limits) ensuring the business logic is entirely transparent and auditable by supply chain managers.

3. **Model Explainability:** Implemented SHAP value visualization, refusing to deploy opaque neural networks, ensuring that marketing managers understand *why* an account is flagged as high-risk.

Challenges Faced

Problem	Root Cause	Solution	Outcome
Negative Sales Values	Retailers process returns and bad debt via negative ledger entries.	Implemented a dedicated transaction mapping function to net out negative quantities against original purchases.	Cleaned monetary features perfectly reflected true cash flow.
Daily Forecast Noise	Forecasting retail sales on a daily basis resulted in massive spikes and 0-value days (weekends/shipping halts).	Resampled the dataset to weekly ('W-MON') frequency and utilized rolling 4-week lags.	Dropped validation MAPE significantly down to 11.81%.
KMeans Density Issues	High monetary outliers were skewing centroid assignments, pushing 99% of customers into one block.	Identified and systematically isolated massive wholesale cancellation outliers representing ~2% of the base.	Resulted in 3 distinct, highly actionable behavioral clusters.

Deployment & Operations ## Current Implementation The RetailPulse platform is currently deployed as a **Local Streamlit Application**. - **Architecture:** Runs securely on `localhost:8501`. - **State Management:** Session-state token authentication restricts unauthorized access to predictive data. - **Data Layer:** Operates entirely on an in-memory Pandas caching mechanism utilizing precomputed CSV and JSON flat files. This negates the overhead of a dedicated SQL server during the pilot validation phase and ensures rapid dashboard rendering. ## Implemented MLOps Practices - **Asset Serialization:** Machine learning models are securely serialized into `.pkl` formats and stored centrally within their respective module boundaries. - **Model Registry Dashboard:** The system features a native UI registry that actively validates the presence of model binaries, logs dataset dependencies, and audits the environment upon startup. - **Decoupled Pipelines:** Model training scripts are physically and logically separated from dashboard inference logic, allowing safe retraining without UI downtime.

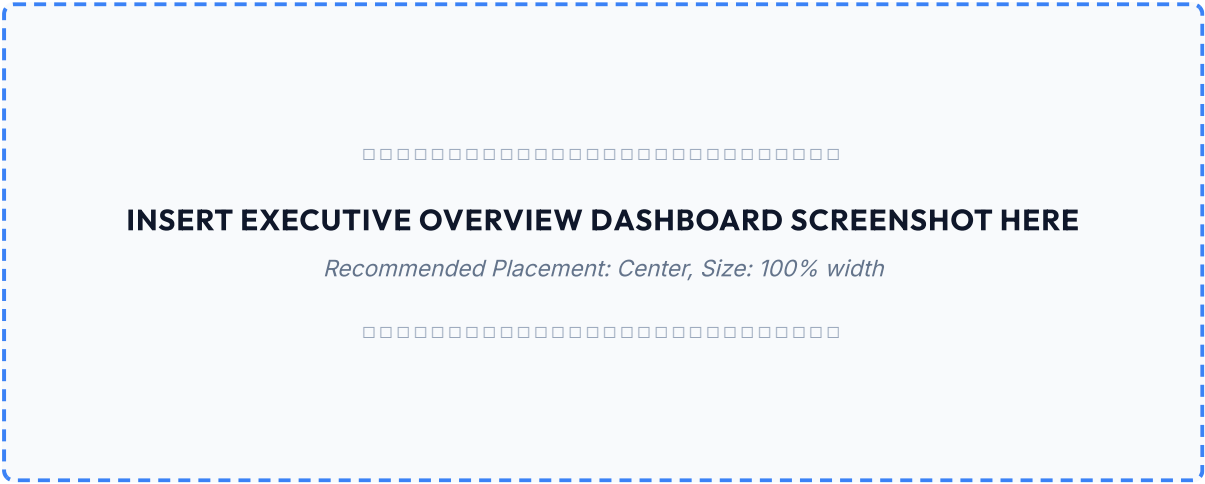
Code & Repository Overview The GitHub repository is architected strictly adhering to industry software engineering standards, ensuring high maintainability and collaboration readiness. -

- ****README Documentation:**** Highly detailed, visually professional markdown complete with Mermaid architecture diagrams, dynamic shields.io badges, and academic disclaimers.
- ****Dependency Management:**** A highly organized `requirements.txt` maps all execution dependencies (Pandas, Scikit-Learn, LightGBM, Streamlit, etc.) required to rebuild the environment.
- ****Code Hygiene:**** - Adherence to standard Python conventions. - A comprehensive repository cleanup was conducted to systematically strip all system cache files (`\_\_pycache\_\_`, `.pyc`), temporary development scripts, and empty folders from the production environment.
- ****Data Artifact Preservation:**** Raw data, cleaned baseline datasets, and predictions are strictly tracked under `processed\_data/` and module-specific `datasets/` boundaries to maintain perfect reproducibility for reporting.

Visual Gallery *The following exhibits demonstrate the actual UI of the implemented platform. For optimal viewing, screenshots should be captured at 1920×1080 resolution and horizontally centered.*

FIGURE 2

Executive Overview Dashboard



Caption: The Executive Overview module providing a high-level index of revenue trends, churn risk distributions, and systemic inventory health.

FIGURE 3

Demand Forecasting Dashboard



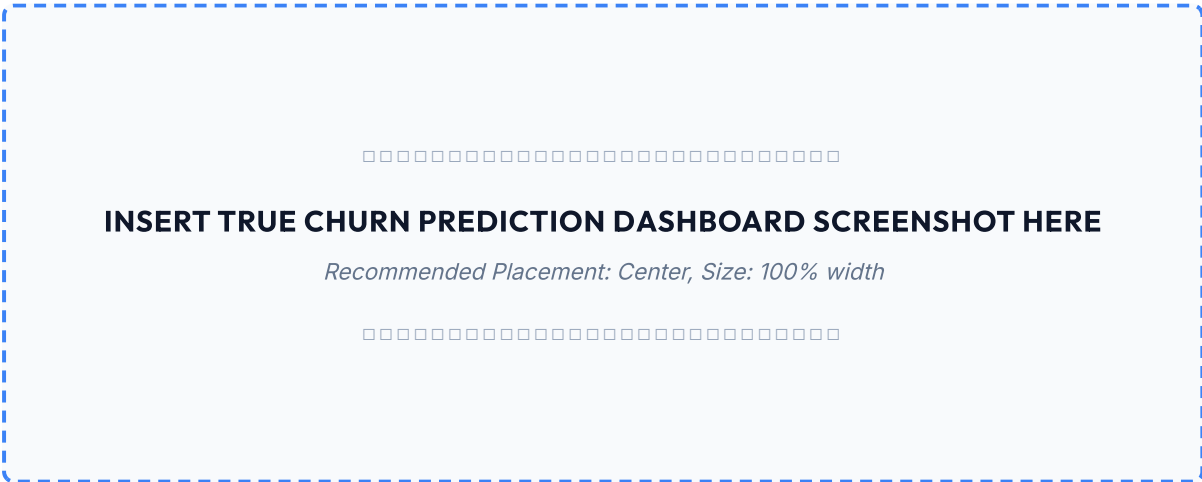
Caption: The 8-week out-of-sample sales projection, tracking validation metrics and holiday-driven demand trends using XGBoost.

FIGURE 4
Customer Segmentation Dashboard



Caption: Interactive breakdown of the KMeans behavioral cohorts, featuring 3D RFM profiling and strategic marketing playbooks.

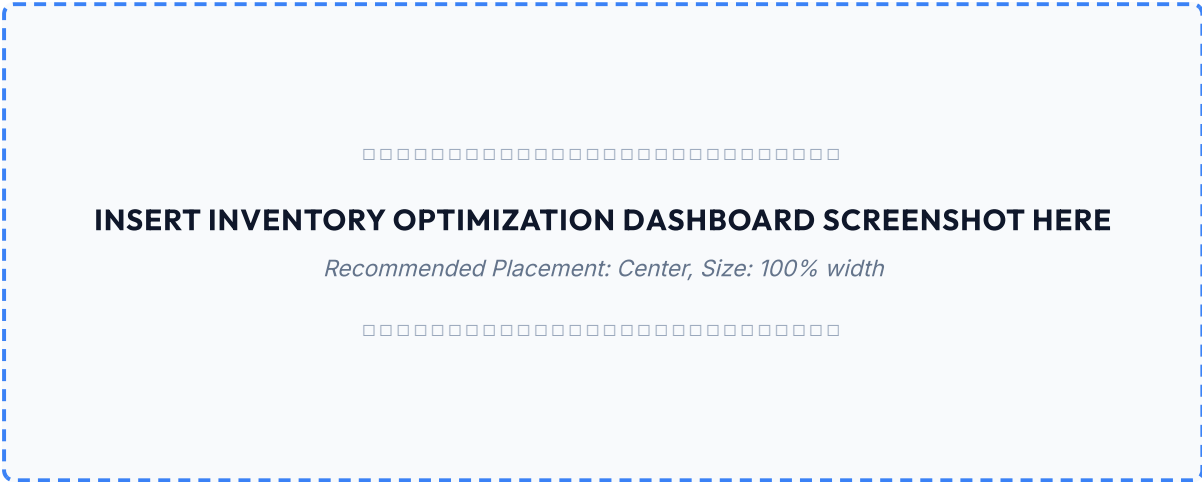
FIGURE 5
True Churn Prediction Dashboard



Caption: Highlighting accounts with a >70% probability of churn over the next 90 days, with explainability provided by SHAP values.

FIGURE 6

Inventory Optimization Dashboard



Caption: ABC catalog categorization alongside calculated Safety Stock, ROP, and EOQ metrics designed to prevent stockouts.

FIGURE 7

Alerts & Monitoring Dashboard



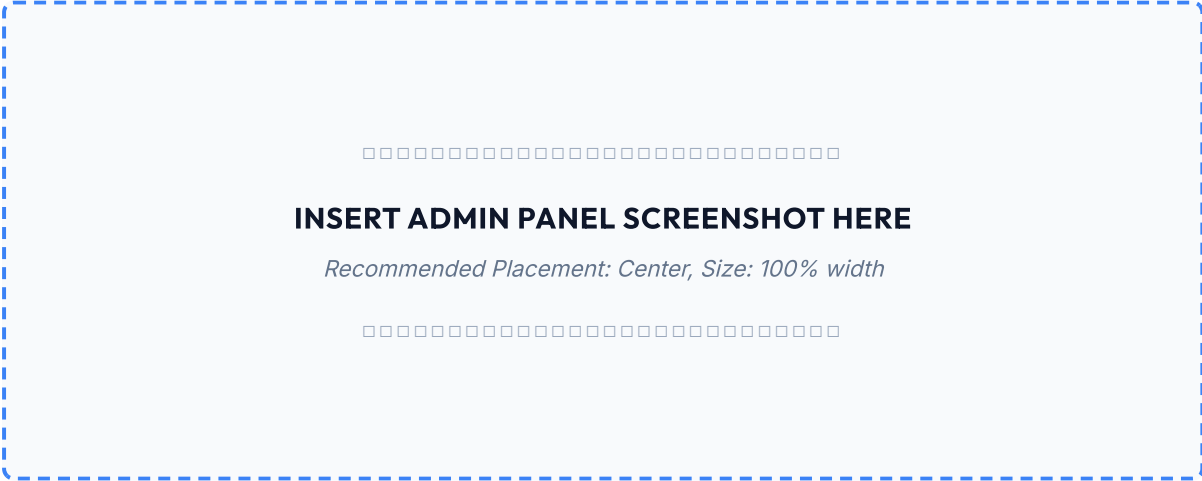
Caption: A centralized warning hub that automatically surfaces critical inventory deficits and urgent revenue risks requiring immediate action.

FIGURE 8
Export Center



Caption: The centralized data export suite allowing authorized personnel to download generated intelligence into CSV and Excel formats.

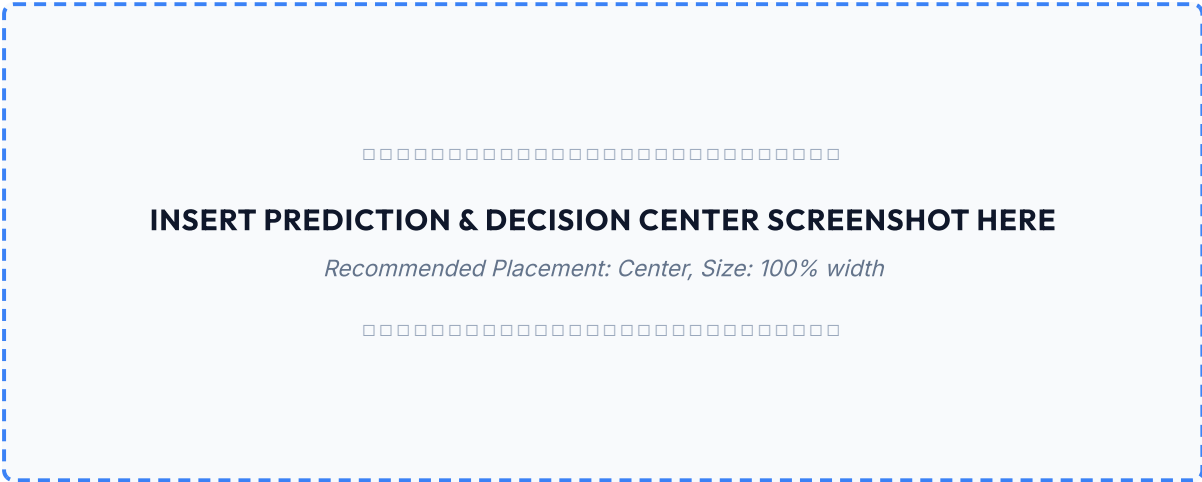
FIGURE 9
Admin Panel



Caption: System configuration panel used by administrators to modify global thresholds (Lead Times, Holding Costs, Service Levels).

FIGURE 10

Prediction & Decision Center



Caption: An interactive query portal providing prescriptive, targeted recommendations for any specific Customer ID or Product ID.

FIGURE 11

Model Registry



Caption: The ML asset management page documenting the status, parameters, and metadata of all serialized model artifacts.

Personal Reflection Developing RetailPulse served as a massive accelerator for my applied software engineering and data science capabilities. - **Business Alignment:** I learned that an algorithm is entirely useless if the business cannot interpret it. Transitioning from Jupyter notebooks to a full-fledged interactive dashboard forced me to consider User Experience (UX) and interpretability (using SHAP). - **Engineering Rigor:** Organizing a repository of this scale taught me the absolute necessity of modular folder structures, decoupled data pipelines, and strict separation of concerns between training scripts and inference scripts. - **Mathematical Validation:** Combining operations research (EOQ/ROP) with machine learning (XGBoost) demonstrated that real-world problems often require hybrid mathematical approaches rather than throwing deep learning at every issue.

Future Enhancements While the pilot platform is completely functional, the following roadmap details planned enhancements for enterprise-grade production:

- **Relational Database Migration:** Transitioning the flat-file CSV architecture to a cloud-managed PostgreSQL instance for real-time concurrent querying.
- **Dockerization:** Containerizing the Streamlit application and ML backend into Docker images for platform-agnostic deployment.
- **Cloud Infrastructure:** Migrating the local deployment to AWS Elastic Beanstalk or Google Cloud Run.
- **REST APIs:** Wrapping the `churn\_prediction\_true` and `Demand\_Forecasting` models in FastAPI endpoints to allow integration with external ERP systems (e.g., Salesforce, SAP).
- **Automated MLOps:** Integrating MLflow for continuous parameter tracking and setting up GitHub Actions (CI/CD) to automatically retrain the models periodically as new transaction data arrives.

Conclusion The RetailPulse AI-Powered Decision Intelligence Platform successfully achieves its core mandate: bridging the gap between raw ledger data and proactive business strategy. By achieving highly validated model metrics (11.81% forecasting MAPE, 71.77% Churn AUC) and coupling them with transparent inventory mathematics, the project proves that advanced machine learning can be effectively democratized for non-technical retail operators. The platform stands as a complete, robust, and portfolio-ready software engineering project demonstrating full-stack data science capabilities—from initial data cleansing and model architecture design to UI development and final operational deployment.

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Appendix ## Appendix A: Hardware & Environment Specifications - **Operating System:** Windows 10/11 / macOS / Linux - **Python Environment:** Conda or standard `venv` - **Minimum RAM Requirement:** 8GB (for running memory-heavy Streamlit local caches and Pandas dataframes) ## Appendix B: Key Mathematical Formulations Used - **Economic Order Quantity (EOQ):** $\sqrt{(2 \times D \times S) / H}$ - **Silhouette Coefficient (Segmentation):** $\frac{b - a}{\max(a, b)}$ - *Where `a` is the mean distance between a sample and all other points in the same class, and `b` is the mean distance between a sample and all other points in the next nearest cluster.*